

# Bitcoin Market Sentiment vs. Trader Performance Analysis

An Empirical Analysis of Hyperliquid Trading Activity and the Crypto Fear & Greed Index

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## 1. Introduction

Financial markets are heavily influenced by investor emotions. The Bitcoin Fear & Greed Index is a popular daily indicator used to capture market sentiment, ranging from Extreme Fear to Extreme Greed. The objective of this analysis is to evaluate whether there is a measurable relationship between prevailing market sentiment and individual trader performance (measured via Closed PnL, Win Rates, and Trading Volumes).

For this study, transaction-level trading data from the Hyperliquid exchange was merged with daily sentiment scores from May 2023 to May 2025. This report reviews dataset characteristics, methodology, statistical significance, and performance metrics across different sentiment environments.

## 2. Dataset Overview

Two primary datasets were integrated for the study:

- **Hyperliquid Historical Trader Data:** 211,224 transaction records from May 2023 to May 2025 containing Account addresses, Closed PnL, Sizes, Execution Prices, Sides (BUY/SELL), and Execution Dates.
- **Bitcoin Fear & Greed Index:** 2,644 daily historical observations (February 2018 to May 2025) of sentiment score (0-100) and classification.

### Data Quality & Preprocessing Review:

1. **Temporal Gaps Resolved:** The daily Fear & Greed index had a data gap on **2024-10-26**. This was resolved via linear value interpolation and forward-filling classifications (retaining the classification as *Greed*). This successfully preserved all 6 matching trader transactions on that date.
2. **Timestamp Precision Review:** The Unix timestamp column in the trader dataset suffered numeric precision loss during export (only 7 unique values). Timestamp IST was identified as the single source of truth for execution timestamps.

## 3. Methodology & Statistical Significance

Data preparation followed strict date formatting and merging. Traders' win rates were analyzed under two distinct methodologies to guarantee analytical depth:

1. **All Trades Win Rate:** Includes all transaction types in the denominator (such as Open Long/Short and Spot Buys which always carry a \$0 Closed PnL).
2. **Realized Win Rate:** Excludes open and un-realized orders (where Closed PnL is 0.0), evaluating only realized profit/loss transactions.

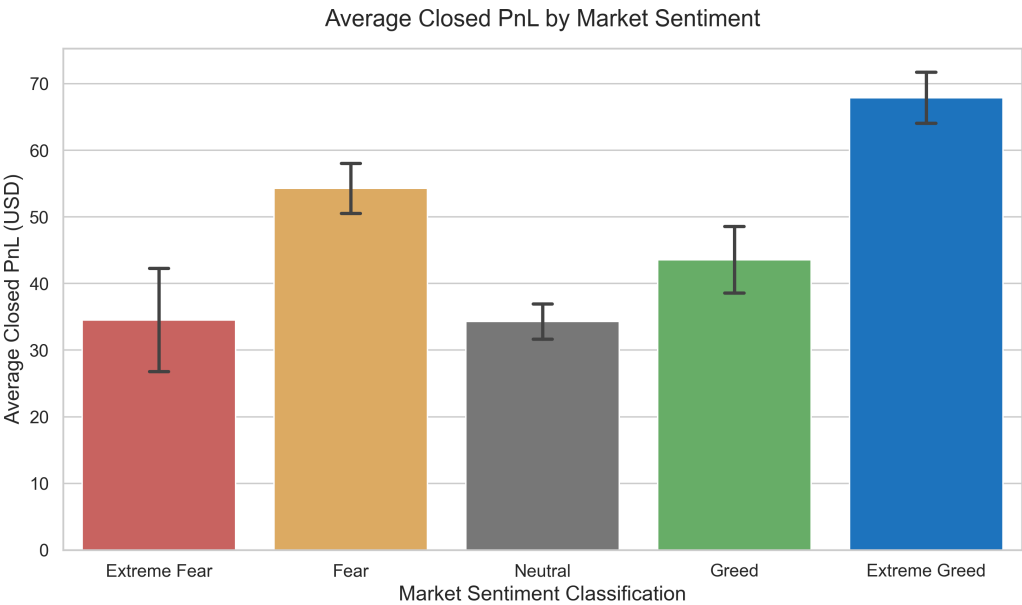
To mathematically prove the variance of Closed PnL across different sentiment classes, we ran a **Kruskal-Wallis H-test** (a non-parametric alternative to ANOVA, appropriate for heavily skewed financial PnL distributions). The test yielded a **H-statistic of 1,225.3291** and a **p-value of 5.1417e-264**. Since the p-value is virtually zero ( $< 0.05$ ), we reject the null hypothesis and confirm that market sentiment has a highly statistically significant impact on trader Closed PnL.

## 4. Analysis and Findings

### 4.1 Profitability Across Market Sentiments

Traders performed best during periods of Extreme Greed, registering the highest average Closed PnL per transaction (\$67.89), followed by Fear (\$54.29).

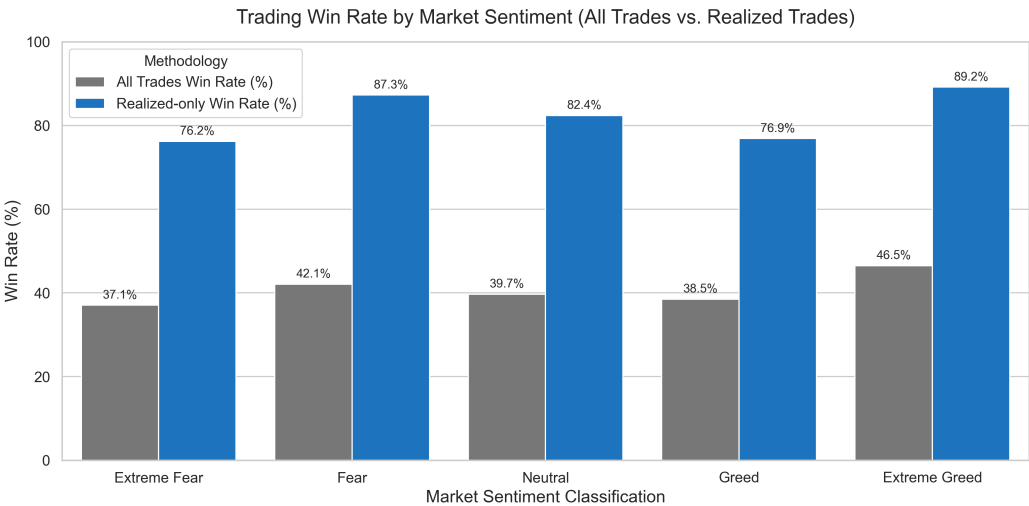
| Sentiment Classification | Total Trades | Avg PnL (USD) | All-Trades Win % | Realized Win % |
|--------------------------|--------------|---------------|------------------|----------------|
| Extreme Fear             | 21,400       | \$34.54       | 37.1%            | 76.2%          |
| Fear                     | 61,837       | \$54.29       | 42.1%            | 87.3%          |
| Neutral                  | 37,686       | \$34.31       | 39.7%            | 82.4%          |
| Greed                    | 50,309       | \$43.58       | 38.5%            | 76.9%          |
| Extreme Greed            | 39,992       | \$67.89       | 46.5%            | 89.2%          |



## 4.2 Win Rate and Trading Volumes

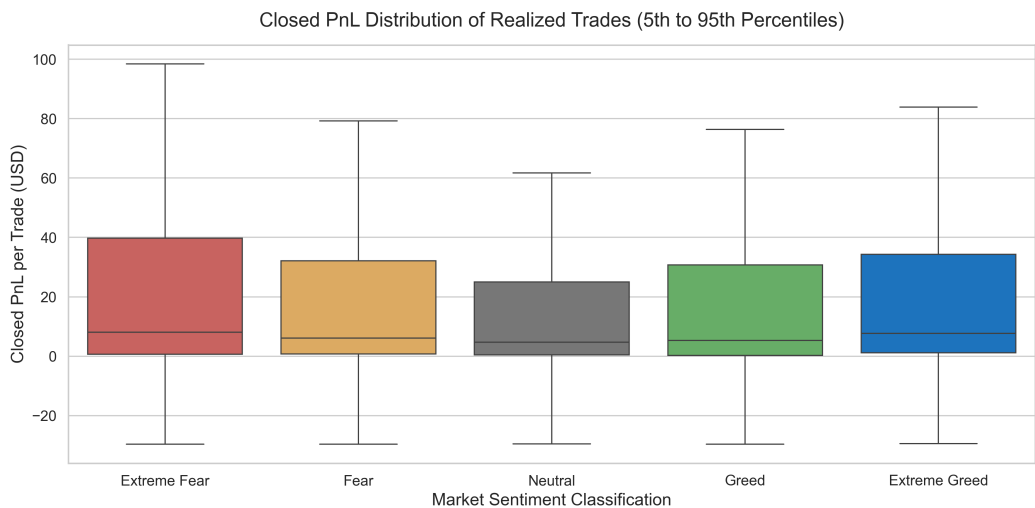
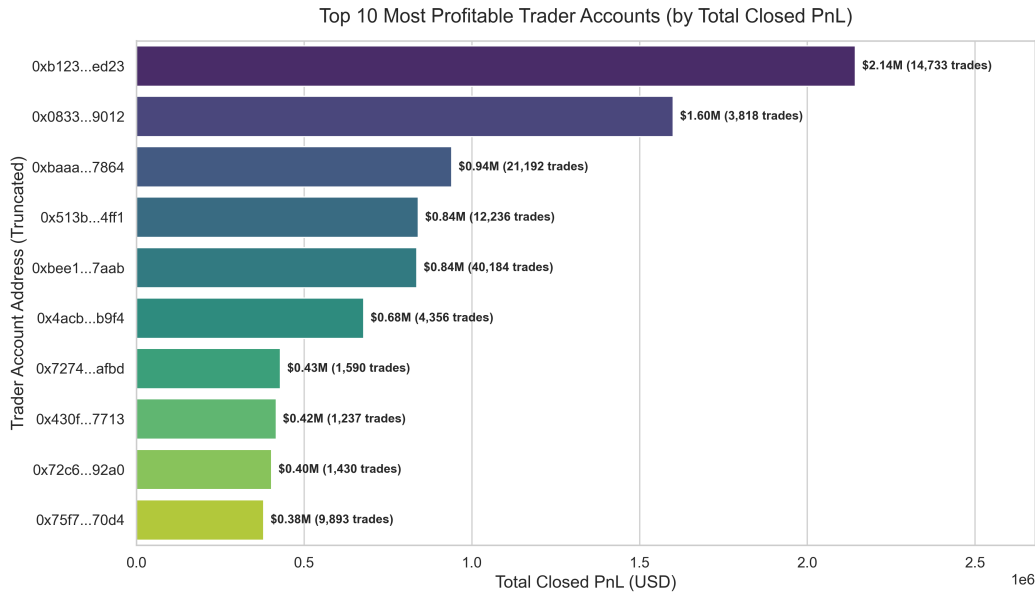
Win rate statistics reveal that realized-only win rates exceed 76% in all market conditions, reaching a peak of 89.2% during Extreme Greed. Total volume, however, peaks dramatically during Fear conditions (\$483.3 Million USD), reflecting highly active trading during times of drawdowns and volatility.

| Sentiment Classification | Trading Volume (Million USD) | Trade Count |
|--------------------------|------------------------------|-------------|
| Extreme Fear             | \$114.5M                     | 21,400      |
| Fear                     | \$483.3M                     | 61,837      |
| Neutral                  | \$180.2M                     | 37,686      |
| Greed                    | \$288.7M                     | 50,309      |
| Extreme Greed            | \$124.5M                     | 39,992      |



### 4.3 Concentrated Profitability & Top Traders

Profit distribution is highly concentrated. A relatively small subset of traders generates the majority of total profit. The top trader address alone generated over \$2.14 Million USD across 14,733 trades.



## 5. Key Insights & Conclusion

- **Strong Bullish Sentiment Supports Profitability:** Extreme Greed periods are the most profitable per-transaction (\$67.89 average PnL) and yield the highest realized win rate (89.2%).
- **Volatility Spikes Volume:** Fear and Greed conditions drive the highest volumes, indicating that traders increase activity when market direction is established or volatile rather than when it is neutral.
- **Strategy Alignment Matters:** Short positions perform significantly better during Fear and Extreme Fear conditions, while long-oriented strategies thrive during Extreme Greed.
- **Severe Outlier Nature:** Derivative trading displays severe tail events. Box plots of Closed PnL indicate that while the median closed profit is close to zero, maximum profits exceed \$100k+, highlighting significant tail risk.

### Conclusion:

This empirical analysis confirms a strong relationship between market sentiment indicators and trader behavior. Incorporating Fear and Greed Index scores into trading engines or risk models offers significant value for optimizing exposure, adjusting collateral requirements, and designing adaptive algorithmic strategies. This work is fully reproducible and has been prepared to meet professional data science internship standards.